Therefore $\frac{a^3r^{15}}{a^3r^{12}} = \frac{32768}{4096}$. Hence $r^3 = 8$. This implies that $r = 2$.

$a^3r^{12} = 4096$ we have $a^3 = 1$. Therefore $a = 1$.

The sum of the firs 8 terms is $\frac{a(1-r^8)}{1-r} = \frac{1-2^8}{1-2} = 255$.

Harmonic Mean

The harmonic mean of a set of positive numbers is the reciprocal of the arithmetic mean of the reciprocals of the set of numbers. That is, if $h_1, h_2, \ldots, h_n$ are positive numbers, then their reciprocals are $\frac{1}{h_1}, \frac{1}{h_2}, \ldots, \frac{1}{h_n}$; the arithmetic mean of the reciprocals is

$$\frac{\frac{1}{h_1} + \frac{1}{h_2} + \cdots + \frac{1}{h_n}}{n}$$

and the reciprocal of this arithmetic mean, that is the harmonic mean of the numbers $h_1, h_2, \ldots, h_n$ is

$$\frac{n}{\frac{1}{h_1} + \frac{1}{h_2} + \cdots + \frac{1}{h_n}}.$$

Definition 5.5

The harmonic mean of a set $\{h_1, h_2, \ldots, h_n\}$ of positive numbers is define as

$$\frac{n}{\frac{1}{h_1} + \frac{1}{h_2} + \cdots + \frac{1}{h_n}}.$$

In particular, the harmonic mean of two positive numbers a and b is $\frac{2}{\frac{1}{a} + \frac{1}{b}}$ which is equal to $\frac{2ab}{a+b}$.

It can be proved that "*For any set of n positive numbers, the geometric mean is greater than or equal to the harmonic mean*". That is, $GM \geq HM$.

Let us prove this inequality $GM \geq HM$ for two non-negative numbers.

Theorem 5.3: If GM and HM denote the geometric mean and the harmonic mean of two non-negative numbers, then $GM \geq HM$. The equality holds if and only if the two numbers are equal.

Proof. Let a and b be any two positive numbers. Then

$$GM = \sqrt{ab} \quad \text{and} \quad HM = \frac{2ab}{a+b}.$$

$$\begin{aligned} GM - HM &= \sqrt{ab} - \frac{2ab}{a+b} \\ &= \frac{\sqrt{ab}(a+b) - 2ab}{a+b} \\ &= \frac{\sqrt{ab}((a+b) - 2\sqrt{ab})}{a+b} \\ &= \frac{\sqrt{ab}(\sqrt{a} - \sqrt{b})^2}{a+b} \\ &\geq 0 \end{aligned}$$

$$GM = HM \Leftrightarrow \frac{\sqrt{ab}\ (\sqrt{a} - \sqrt{b})^2}{a+b} = 0$$

$$\Leftrightarrow \sqrt{ab}\ (\sqrt{a} - \sqrt{b})^2 = 0$$

$$\Leftrightarrow a = b$$

Thus $GM - HM \geq 0$ and hence $GM \geq HM$. ■